Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Sachchit Shankar	sachshan@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to turn telemetry data into an organize graph, then put that graph into Drafter.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Currently, you can use NumPy, Matplotlib and drafter to incorporate this. NumPy can organize and calculate the information from the distance measurements , Matplotlib can turn the measurements into a graph, and then Drafter can display the graph on the website. These tools will be working together to make it easy to understand how distance changes over time!

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

Essentially, for me, I would hook up an arduino to an ultrasonic sensor, the sensor would

sense objects for 10 seconds then report their relevant distances, put them into an array, then put the time from 1-10 in an array, then I would use Matplotlib and NumPy to graph all that, then I would put all of that into a drafter website, where it shows the latest image and keeps updating.

However the TA will not have access to the data, so what I will do is just use two arrays at random. Imagine a list of arrays, and I use the import random to choose two arrays, and from there it will graph and put the graph on Drafter. They would click the button on drafter and it will show them the updated graph, if there is no later graph, it would say that is the latest.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Or even, the user can input distance values, and a set amount of time, both using float values, and then it would graph those. The progrma will then use the data to create a graph of how distance changes over time.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The inputs would be time values, integers or floats and distance measurements, stored in arrays. For my actual project, the Arduino and ultrasonic sensor will collect the distance measurements, while I will use preset arrays of sample data for the TAs since they might not have an arduino. The time values will correspond to each distance value.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

First I would organize the time and distance measurements in NumPY. Then I will use Matplotlib to graph the data and calculate the statistics like the average and line of the best fit. It uses matplotlib to graph, then NumPy for averages, line of best fit, et cetera, and then I'll try and figure out a way to show an R^2 value? Just how we do for our chem lab reports.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The output would be a graph showing the distance over time, displaying on a website using Drafter. It could ALSO show statistics such as the average distance and even possibly the R^2 value (as mentioned earlier, we use R^2 in lab reports for chemistry).

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Time, Distance in Centimeters, Arrays

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

For the most important thing would be the distance in centimeters, I'd have to calculate the distance by using the time it takes for the ultrasonic sensor to send its signal and then receive it, then convert that distance into centimeters.

Details I'll need would be measurement's time, distance, measurement order, and the number of readings.
And then irrelevant details would be like object color and material.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Wiring the Arduino correctly, and then actually making sure I know how to use Matplotlib and NumPy. I have been learning recently, but I am not proficient at it. And then making sure I properly format the data into a graph, and even make sure I input it all correctly

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I most likely will NOT use an LLM for this, other than for the Drafter part because I am not that confident. I would ask what each line does, make sure I understand it, and actually go line by line, I would code the line and ask if that is correct, if not, please explain. I would never ask it to code the entire thing, because that takes away from my learning.

I can work on this project on Saturdays and Sundays, for a couple hours each. Those work best, since my arduino kit is at home.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

The reason I am using an Arduino is because I want to test how well I know hardware and software. This is one step into becoming a mechanical engineer. Coding alone won't be enough, if I can't actually apply it to physical systems. I think this would also look good as a personal project, and eventually I want to use like sonar mapping, the ultrasonic sensor uses sounds to map out a room visually, though this is very complex. I was inspired by like Beluga whales, and other animals that use EchoLocation. It is a cool concept, and a bit tricky for a beginner in circuits and code, but I know this will help me definitely. It is something I am passionate about, and it will also allow me to learn Drafter more in depth, because I'll be adding more images, and more routes to update pages too. I will also make the Drafter website have a full description of the project, almost like a project portfolio, with many sub-pages also, which will allow me to "showcase" what I've built, while simultaneously learning Drafter and coding more in-depth.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

> *I want to build a website that displays distance measurements from an Arduino ultrasonic sensor over time. What are three challenges I might face, and what are three simple features I could include in my project?*

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

> *AI said that some challenges would be connecting the Arduino to the website, displaying measurements over time and handling inaccurate sensor readings . It did however suggest features like showing the current distance keeping history of measurements , and calculating the minimum and maximum distances, the biggest challenge would probably be getting the Arduino's measurements into the website reliably.*

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

> *Connecting the arduino directly to the website and getting the live measurements might be too complicated because I have to figure out how to transfer data between the hardware and the website. A simpler alternative would be to record the distance measurements first, and then use like a CSV file to enter them manually into the website, I could still use NumPY for calculating statistics, Matplotlib to create a graph , and Drafter to display everything on the website.*

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project idea became much more focused because I did realize that connecting the Arduino directly to the website might be a bit too complicated for this class, but I will try it, and if not, I can just stick to using simple arrays and then using matplotlib and NumPy, but I will try it.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

Drafter Documentation, Arduino Documentation, YouTube, and a bit of CHATGPT for clarification.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I would read the error, and then inspect variable types and arrays, test the individual parts , consult the documentation and then re-run the program.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a website that displays distance measurements from an Arduino ultrasonic sensor over time. What are three challenges I might face, and what are three simple features I could include in my project?

For your Arduino ultrasonic sensor website, I would focus on challenges that are realistic for a beginner programming project and features that are simple to implement.

1. Three Challenges You Might Face

Challenge	Why it's difficult	Simple solution
1. Connecting Arduino to the website	The Arduino measures distance, but the website needs a way to receive that data.	Use a USB connection and a small Python program to send readings to the website.
2. Displaying measurements over time	You need to update the displayed distance whenever a new measurement arrives.	Store each reading with a timestamp and display the latest measurement.
3. Handling inaccurate readings	Ultrasonic sensors can sometimes return inconsistent measurements or fail to detect an object.	Ignore invalid readings and display a message when a measurement is unavailable.

2. Three Simple Features to Include

Feature	What it does	Why it's useful
1. Live Distance Display	Shows the most recent distance measured by the sensor in centimeters.	Lets users see how far away an object is without checking the Arduino's Serial Monitor.

2. Distance History	Displays previous measurements alongside their timestamps.	Lets users observe how the distance changes over time.
3. Minimum and Maximum Distance	Shows the smallest and largest valid distances recorded during a session.	Helps users understand the range of distances measured.

3. What the Website Could Look Like

For example, your website could display:

- Current Distance: 24.5 cm
- Minimum Distance: 12.3 cm
- Maximum Distance: 68.7 cm
- Distance History: A simple table showing previous measurements and their timestamps.

You could build the website using basic HTML, CSS, and JavaScript, with Python handling communication between the Arduino and the website.

4. My Recommendation for a Beginner

Start with just these three things:

1. Live distance display — show the latest sensor reading.
2. Measurement history — list the previous readings with timestamps.
3. Minimum and maximum distance — calculate the smallest and largest readings.

You don't need a complicated dashboard, user accounts, or a database for the initial version.

The biggest technical challenge will probably be getting the Arduino's measurements into the website reliably. Once that connection works, the other features are much easier to build.

I think connecting the Arduino directly to the website and getting live measurements might be too complicated for an introductory programming project because I would have to figure out how to transfer data between the hardware and the website. So like what is a simpler alternative that would still let me graph distance measurements over time using NumPy, Matplotlib, and Drafter?

A simpler alternative is to record distance measurements first, then upload or enter them into your website to graph them over time. You can still use NumPy, Matplotlib, and Drafter without having to connect the Arduino directly to your website.

1. Recommended Approach: Record, Then Graph

Here's how your project would work:

1. Measure: Use your Arduino and ultrasonic sensor to measure distances to an object.
2. Record: Save the measurements in a CSV file or enter them manually into your website.
3. Process: Use NumPy to calculate statistics, such as the average, minimum, and maximum distance.
4. Visualize: Use Matplotlib to create a line graph showing how the distance changes over time.
5. Display: Use Drafter to build a website where users can view the graph and measurement statistics.

For example, you could move an object closer to and farther from your ultrasonic sensor, recording a measurement every second.

Your data might look like this:

Time (seconds)	Distance (cm)
0	50
1	45
2	38
3	30
4	25
5	35
6	48

Your website would graph these measurements to show how the distance changes over time.

2. Three Simple Features

I'd recommend these features for an introductory programming project.

Feature 1: Distance-over-Time Graph

Use Matplotlib to create a line graph.

- X-axis: Time in seconds.
- Y-axis: Distance in centimeters.
- Purpose: Show how the distance changes as an object moves.

This is the main feature of your project.

Feature 2: Distance Statistics

Use NumPy to calculate three statistics from the measurements.

- Average distance.
- Minimum distance.
- Maximum distance.

For example, if your measurements are `[50, 45, 38, 30, 25, 35, 48]`, NumPy can calculate these values without requiring you to write the calculations yourself.

Feature 3: User-Entered Measurements

Create a simple form in Drafter where users can enter their own distance measurements.

For example, a user could enter `42.5` centimeters and add it to the dataset.

The website could then update the graph and statistics using the new data.

This makes your project interactive without requiring live hardware communication.

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3. How the Libraries Fit Together

Technology	Role in your project
Drafter	Creates the website, input forms, buttons, and results page.
NumPy	Calculates average, minimum, and maximum distances.
Matplotlib	Generates the distance-over-time graph.
Arduino	Measures the actual distance using the ultrasonic sensor.
CSV file	Stores measurements so your program can access them later.

You wouldn't need a database, a server that communicates with the Arduino, or complicated networking.

4. One Important Design Decision

You have two good options for collecting the measurements:

Option A — Use actual Arduino measurements (recommended)

Record measurements using your Arduino and save them to a CSV file. Your Python program reads that file and graphs the data.

Option B — Enter measurements manually

Let users enter distances through the website. This is simpler because you can focus on Python, graphing, and website development rather than hardware integration.

You could even start with Option B and add CSV importing later.

My recommendation

Build a Distance Tracking and Visualization Website with these three features:

1. A line graph showing distance over time.
2. Statistics showing average, minimum, and maximum distances.
3. A form for entering new measurements.

This gives you a manageable introductory programming project that combines NumPy, Matplotlib, and Drafter while still relating to your Arduino hardware interests.

The key is that your project focuses on analyzing and visualizing sensor data rather than building a complicated connection between hardware and a website.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.